

Supplement to
Modeling the Modeler

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1 EXAMPLES

In order to better understand the role of concordance and the structure that it imposes, we provide three examples of concordant experimental paradigms. These examples also highlight the flexibility of our framework at capturing disparate, and empirically relevant, experimental protocols.

1.1 EXAMPLE: EXPECTED UTILITY

Example 1. Consider a modeler who believes that the subject has expected utility preferences and can ask the subject to make only a single choice from one menu. In this case, the set of choice patterns $\mathbb{Z}$ is simply the ℓ -dimensional simplex of all relevant lotteries, denoted by Δ and the set of experiments is $\mathbb{E}^\Delta = \{(A, \mathcal{P}) \mid A \subseteq \Delta \text{ is finite, } \mathcal{P} \text{ partitions } A\}$.

The set of types is in one-to-one correspondence with the set of normalized expected utility functions over Δ . Thus,

$$\Theta^\Delta = \{\theta \in \mathbb{R}^\ell \mid \theta_\ell = 0\}$$

and the choice function for each $\theta \in \Theta$ is determined by utility maximization:¹

$$c_\theta(A) \in \arg \max_{p \in A} \sum_{i=1}^{\ell} \theta_i p_i.$$

Let Ω^Δ be the smallest algebra on Θ^Δ containing $W_{A,B}$ for all finite $A \subseteq \Delta$ and $B \subseteq A$, and $\mu \in \mathbb{P}(\Theta^\Delta, \Omega^\Delta)$.

This specialization of our model is closely related to the setup of random expected utility models à la [Gul and Pesendorfer \(2006\)](#) (henceforth, GP) and indeed, the experimenter's prior μ defines a GP random expected utility model.² GP show that a belief μ can be identified from random choice data exactly when it is *regular*, that is, if for each $p, q \in \Delta$,

¹Of course, this ignores how ties are broken; it is possible to expand this environment to allow the modeler to entertain probabilistic beliefs about tie-breaking as well, but this is irrelevant for regular measures, as defined below.

² Ω^Δ coincides with the algebra of utilities defined in GP.

$\mu(\{\theta \in \Theta^\Delta \mid \sum_i \theta_i p_i = \sum_i \theta_i q_i\}) = 0$. Within this expected utility framework, the modeler's theory is concordant with the set of all finite decision problems whenever it can be identified by a random expected utility model.

PROPOSITION 2. *If μ is regular, then $(\Delta, (\Theta^\Delta, \Omega^\Delta, \mu), \mathbb{E}^\Delta)$ is concordant.*

Proof of Proposition 2. For a set $X \subseteq \mathbb{R}^\ell$, let $\text{conv}(X)$, $\text{int}(X)$, and $\text{cl}(X)$ denote the convex hull, the interior, and the closure of X , respectively. If X is convex, then $\text{ext}(X)$ collects all the extreme points of X and $\text{ri}(X)$ denotes the relative interior of X . When it is not confusing to do so, we will write $\text{ri}(X)$ and $\text{ext}(X)$ to mean $\text{ri}(\text{conv}(X))$ and $\text{ext}(\text{conv}(X))$ for non-convex X .

For convex X , let $F \subset X$ be called a face if $\alpha x + (1 - \alpha)y \in F$ (for $x, y \in X$) implies $x, y \in F$. Let $\mathbb{F}(X)$ denote the set of all (non-empty) faces of X and $\mathbb{F}^\circ(X) = \{\text{ri}(F) \mid F \in \mathbb{F}(X)\}$. If $X \subseteq \mathbb{R}^\ell$ is a convex set and $\text{ext}(X)$ is finite then X is called a polytope. Let poly denote the set of all polytopes in $\mathbb{R}^\ell$. If $X \in \text{poly}$, then $\mathbb{F}(X)$ is finite.

If $K \subseteq \mathbb{R}^\ell$ and $\lambda K \subseteq K$ for all $\lambda \geq 0$ then K is called a *cone*. We say a cone K is generated by X if $K = \{\lambda x \mid x \in X, \lambda \geq 0\}$. A cone K is *polyhedral* if it is generated by a polytope; let $\mathcal{K}^*$ denote all such cones. Let $\mathcal{K}$ denote the set of pointed polyhedral cones, those cones with $\mathbf{0} \in \text{ext}(K)$. The face of a polyhedral cone is a polyhedral cone.

For $X \in \text{poly}$, let

$$X^* = \bigcup_{I \subseteq \text{ext}(X)} \sum_{i \in I} \frac{i}{|I|}$$

The set X^* is a set that contains one point in the relative interior of every face of X . Further, given a partition of $\mathcal{H} = \{H_1 \dots H_n\}$ of $\mathbb{F}(X)$, let $\mathcal{H}^* = \{H_1^* \dots H_n^*\}$ denote the partition of X^* defined via $H_i^* = X^* \cap (\bigcup_{F \in H_i} \text{ri}(F))$.

For $X \subseteq \mathbb{R}^\ell$ (convex or not) and $x \in X$ let $N(X, x) = \{u \in \mathbb{R}^\ell \mid \langle u, (y - x) \rangle \leq 0, \text{ for all } y \in X\}$ denote the normal cone of X at x . Notice that under the experimental paradigm $(\Delta, (\Theta^\Delta, \Omega^\Delta, \mu), \mathbb{E}^\Delta)$,

$$W_{A,B} = \bigcup_{x \in B} N(A, x)$$

for all finite $A, B \subseteq \Delta$.

For $X \in \text{poly}$, and a face $F \in \mathbb{F}(X)$, let $N(X, F) = \bigcap_{x \in F} N(X, x)$. It follows that $N(X, F) = \{u \in \mathbb{R}^\ell \mid F \subseteq \arg \max_{x \in D} \langle u, x \rangle\} = N(X, x)$ for any $x \in \text{ri}(F)$. Notice therefore that

$$\bigcup_{x \in H_i^*} \text{ri}(N(X^*, x)) = \bigcup_{F \in H_i} \text{ri}(N(X, F)). \quad (1)$$

It is immediate that $N(X, F)$ is closed and in $\mathcal{K}^*$. Let $\mathcal{N}(X) = \{N(X, F) \mid F \in \mathbb{F}(X)\}$ denote the normal fan of X ; that $\mathcal{N}(X)$ is a fan indicates that it is a family of cones with the following two properties:

- (i) Every nonempty face of a cone in $\mathcal{N}(X)$ is also a cone in $\mathcal{N}(X)$.
- (ii) The intersection of any two cones in $\mathcal{N}(X)$ is a face of both.

Furthermore, $\mathcal{N}(X)$ is complete (the union of $\mathcal{N}(X)$ is Θ^Δ). Let

$$\mathcal{N}^\circ(X) = \{\text{ri}(N(X, F)) \mid F \in \mathbb{F}(X)\}.$$

LEMMA 1. *The following are true for all convex X and Y :*

1. $\text{cl}(\text{ri}(X)) = \text{cl}(X)$ (Theorem 6.3 of [Rockafellar \(1970\)](#)).
2. $\mathbb{F}^\circ(X)$ is a partition of X (Theorem 18.2 of [Rockafellar \(1970\)](#)).
3. Let $F \in \mathbb{F}(X)$ and $Y \subseteq X$ be such that $\text{ri}(Y) \cap F \neq \emptyset$, then $Y \subseteq F$. (Theorem 18.1 of [Rockafellar \(1970\)](#)).

LEMMA 2. *For polytopes X and X' , the following are equivalent:*

- (i) $X = \alpha X' + Z$, for some polytope Z and $\alpha > 0$
- (ii) for all $K \in \mathcal{N}(X)$ there is a $K' \in \mathcal{N}(X')$ such that $K \subseteq K'$
- (iii) $\mathcal{N}^\circ(X)$ refines $\mathcal{N}^\circ(X')$.

Proof. (i) $\leftrightarrow$ (ii) Theorem 15.1.2 of [Grünbaum \(2003\)](#)

(ii) $\rightarrow$ (iii) Take some $\text{ri}(K) \in \mathcal{N}^\circ(X)$. Let $K^\dagger = \bigcap \{K' \in \mathcal{N}(X') \mid K \subseteq K'\}$, which is an element of $\mathcal{N}(X')$ by (ii) and the properties of fans. Moreover, by construction $K \not\subseteq F$ for any $F \in \mathbb{F}(K^\dagger)$ with $F \subsetneq K^\dagger$. Thus, by (the contra-positive of) Lemma 1.3, we have that $\text{ri}(K) \cap \text{ri}(F) \neq \emptyset$ for all such F . Then, since $\mathbb{F}^\circ(K^\dagger)$ partitions $K^\dagger$, it follows that $\text{ri}(K) \subseteq \text{ri}(K^\dagger)$.

(iii) $\rightarrow$ (ii) Take some $K \in \mathcal{N}(X)$. Then by (iii) $\text{ri}(K) \subseteq \text{ri}(K')$ for some $K' \in \mathcal{N}(X')$. Since K and K' are closed, we have $K = \text{cl}(K)$ and $K' = \text{cl}(K')$. Thus, $K = \text{cl}(\text{ri}(K)) \subseteq \text{cl}(\text{ri}(K')) = K'$, where both equalities come from Lemma 1.1 and the inclusion relation from the fact that taking closures is subset preserving. ★

(C1) follows from the definition of Ω^Δ and $\mathbb{E}^\Delta$. We will show (C2): that any partition can be captured up to μ -equivalence. Let $\mathcal{W} = \{W_1, \dots, W_n\} \in \text{part}(\Theta^\Delta)$. First, from [Gul and Pesendorfer \(2006\)](#) Proposition 6 (ii), we can write each $W_i \in \Omega$ as the finite union of elements in $\mathcal{K}$: $W_i = \bigcup_{j=1}^{m_i} \text{ri}(K_i^j)$. Moreover, by [Gul and Pesendorfer \(2006\)](#) Proposition 4, each $K_i^j = N(X_i^j, x_i^j)$ for some polytope X_i^j and $x_i^j \in X_i^j$. Thus $\text{ri}(K_i^j) \in \mathcal{N}^\circ(X_i^j)$.

Let $a = m_1 + \dots + m_n$, where m_i is the number of cones used to represent W_i , and consider the polytope $X = \sum_{i=1}^n \sum_{j=1}^{m_i} \frac{1}{a} X_i^j$. By Lemma 2, $\mathcal{N}^\circ(X)$ refines each $\mathcal{N}^\circ(X_i^j)$. Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a partition of $\mathbb{F}(X)$ defined by

$$H_i = \{F \in \mathbb{F}(X) \mid \text{ri}(N(X, F)) \subseteq W_i\} \quad (2)$$

Now take some $i \leq n$ and $\theta \in W_i$. So, there exists some $j \leq m_i$ such that $\theta \in \text{ri}(K_i^j) \in \mathcal{N}^\circ(X_i^j)$. Since $\mathcal{N}^\circ(X)$ is a partition of Θ^Δ , there exists some $F \in \mathbb{F}(X)$ with $\theta \in \text{ri}(N(X, F))$, and furthermore, since this partition refines $\mathcal{N}^\circ(X_i^j)$, $\text{ri}(N(X, F)) \subseteq \text{ri}(K_i^j) \subseteq W_i$. Hence $F \in H_i$ and so $\theta \in \bigcup_{F \in H_i} \text{ri}(N(X, F))$. We have established that $W_i \subseteq \bigcup_{F \in H_i} \text{ri}(N(X, F))$, and since the other inclusion is obvious, it follows that $W_i = \bigcup_{F \in H_i} \text{ri}(N(X, F))$. Now, on the basis of (1), we have

$$W_i = \bigcup_{x \in H_i^*} \text{ri}(N(X^*, x)) \quad (3)$$

Finally, by Lemma 2 of [Gul and Pesendorfer \(2006\)](#), we know that for any regular μ , it must be that $\mu(\text{ri}(N(X^*, H^*))) = \mu(N(X^*, H^*))$. Thus, $\{W_{X^*, H^*}\}_{H^* \in \mathcal{H}^*} \approx_\mu \mathcal{W}$. ■

We observe that the restriction to offering only a single menu in the previous example is without loss of generality. Although in many environments observing only one-shot behavior is limiting, this is not the case under expected utility. The reason is that for any finite $A, A' \subseteq \Delta$ and linear $u : \Delta \rightarrow \mathbb{R}$,

$$\left. \begin{array}{l} p \in \arg \max_A u(\cdot) \\ q \in \arg \max_{A'} u(\cdot) \end{array} \right\} \text{ if and only if } \alpha p + (1 - \alpha)q \in \arg \max_{\alpha A + (1 - \alpha)A'} u(\cdot)$$

where $\alpha A + (1 - \alpha)A'$ is the weighted Minkowski sum of A, A' .³ Hence, observing a single choice from $\alpha A + (1 - \alpha)A'$ is equivalent to observing a choice from A and A' and thus, it is without loss of generality for the theory to only consider to offer the subject a single menu to make a choice from. □

We will now recast structural invariance in a domain specific manner, illuminating that the concrete notion of invariance that is relevant to the expected utility model is *translation invariance*.

Consider the experiment $(A, \{P_1, \dots, P_n\})$ and some other A' ; notice that the linearity of each u requires that $u \in W_{A, P_i}$ if and only if $u \in W_{\alpha A + (1 - \alpha)A', \alpha P_i + (1 - \alpha)A'}$. Indeed, there must be some $p \in P_i$ that maximizes u over A and $q \in A'$ that maximizes u over A' , so that $\alpha p + (1 - \alpha)q \in \alpha P_i + (1 - \alpha)A'$ maximizes u over the mixture. Hence, translating an experiment by mixing both the decision problem and the observability partition with some common A' does not alter which preference sets can be identified. This particular form of invariance is captured by the following axiom.⁴

(B1) Translation Invariance. For $A, B \in \mathbb{D}$, we have

$$(A, \{P_1, \dots, P_n\}) \sim (\alpha A + (1 - \alpha)B, \{Q_1, \dots, Q_n\}),$$

whenever $Q_i \subseteq (\alpha P_i + (1 - \alpha)B)$ for all $i \leq n$.

Recall that [A3](#) also implies that the value of identification should not depend on zero probability perturbations. The following axiom reflects this implication.

(B2) Belief Consistency. Fix $c \geq 0$, $A \in \mathbb{D}$, and let $\{P_1, P_2, \dots, P_n\}$ be a partition of A such that $\mu(W_{A, P_1}) = 0$. Then:

$$(A, \{P_1, P_2, \dots, P_n\}, c) \sim (A, \{P_1 \cup P_2, \dots, P_n\}, c).$$

These two axioms capture invariance directly in the Expected Utility specialization of

³That is, $\alpha A + (1 - \alpha)A' = \{\alpha p + (1 - \alpha)q \mid p \in A, q \in A'\}$ for all $\alpha \in [0, 1]$ and $A, A' \subseteq \Delta$.

⁴The reason there is a subset, rather than set equality, in the axiom is that it is possible that $r \in \alpha A + (1 - \alpha)A'$ is not a unique mixture of two elements. That is, $r = \alpha p + (1 - \alpha)q = \alpha p' + (1 - \alpha)q'$ for some $p, p' \in A$ and $q, q' \in A'$. For these elements, the cell of the partition in which they reside is not determined, but, it turns out not to matter. See the appendix for the formal argument.

our framework:

PROPOSITION 3. *Let $\succsim$ be defined over Δ . Then $\succsim$ satisfies B1 and B2 iff it satisfies A3*

Proof of Proposition 3. Let $(A, \{P_1, \dots, P_n\})$ and $(B, \{Q_1, \dots, Q_m\})$ be such that $\{W_{A,P_i}\}_{i \leq n}$ is μ -equivalent to $\{W_{B,Q_i}\}_{i \leq m}$. Furthermore, from Lemma 2, we can assume there are $1 \leq k \leq n$ elements of each partition with positive μ -probability and for each $i \leq k$, $\mu(W_{A,P_i}) = \mu(W_{A,P_i} \cap W_{B,Q_i}) = \mu(W_{B,Q_i})$.

Consider the problem $C = \frac{1}{2}A + \frac{1}{2}B$. For each $i \leq n$, define $R_i \subseteq C$ as $R_i = \{\frac{1}{2}P_i + \frac{1}{2}B\} \cap \text{ext}(C)$. Clearly, we have for each $i \leq n$, $R_i \subseteq \frac{1}{2}P_i + \frac{1}{2}B$; it follows from B1 that $(A, \{P_1, \dots, P_n\}) \sim (C, \{R_1, \dots, R_n\})$.

Now for each $i \leq k$, let $R'_i = R_i \cap (\frac{1}{2}P_i + \frac{1}{2}Q_i) \cap \text{ext}(C) = (\frac{1}{2}P_i + \frac{1}{2}Q_i) \cap \text{ext}(C)$. The final equality arises from the fact that each extreme point of C has a unique decomposition as elements of A and B (so that any $x \in (\frac{1}{2}P_i + \frac{1}{2}Q_i) \cap \text{ext}(C)$ was not in R_j for $j < i$). We claim that $\mu(W_{C,R_i \setminus R'_i}) = 0$. Indeed,

$$\begin{aligned} W_{C,R_i \setminus R'_i} &\subseteq W_{A,P_i} \cap \bigcup_{j \neq i} W_{B,Q_j} \\ &= \bigcup_{j \neq i} (W_{A,P_i} \cap W_{B,Q_j}) \\ &\subseteq \bigcup_{j \neq i} ((W_{A,Q_i} \cap W_{B,Q_j}) \cup (W_{A,P_i} \setminus W_{B,Q_i})) \end{aligned}$$

The claim then follows from the fact that for all $i \neq j$, $W_{A,Q_i} \cap W_{B,Q_j} = \emptyset$ and $\mu(W_{A,P_i} \setminus W_{B,Q_i}) = 0$ (from μ -equivalence).

By repeatedly appealing to B2, we can see that

$$(C, \{R_1, \dots, R_n\}) \sim (C, \{R'_1, \dots, R'_k, R^\dagger\}),$$

where $R^\dagger = C \setminus \bigcup_{i \leq k} R'_i$. We make use the fact for $i > k$, $\mu(W_{C,R_i}) = 0$ on account of the fact that $W_{C,R_i} \subseteq W_{A,P_i}$. Thus we have

$$(A, \{P_1, \dots, P_n\}) \sim (C, \{R_1, \dots, R_n\}) \sim (C, \{R'_1, \dots, R'_k, R^\dagger\})$$

A symmetric argument ensures that also $(B, \{Q_1, \dots, Q_m\}) \sim (C, \{R'_1, \dots, R'_k, R^\dagger\})$, and so the two experiments are indifferent, as is required. ■

1.2 EXAMPLE: CHOICE CORRESPONDENCES

Example 2. Consider the textbook exercise of a modeler who wishes to infer a subject's preferences over some set X of cardinality n . She will observe what the subject's admissible choices out of each possible subset of X . Let $\mathcal{M}(X) = \{M_1, \dots, M_{2^n-1}\}$ be the set of all non-empty subsets of X and define the set of all possible choice patterns as

$$\mathbb{Z} = \{\times_{i=1}^{2^n-1} C_i \mid C_i \subseteq M_i \text{ for all } i\}.$$

The interpretation being that C_i is the set that contains all the alternatives the subject considers admissible when facing M_i .

Suppose the modeler's theory is that the subject is maximizing a complete and transitive preference. Let Θ be the set of all weak orders over X (for $\theta \in \Theta$ write $x \succ_{\theta} y$ rather than $x\theta y$) and

$$c_{\theta}(\mathcal{M}(M_1), \dots, \mathcal{M}(M_{2^n-1})) = \times_{i=1}^{2^n-1} \{x \in M_i \mid x \succ_{\theta} y, \text{ for all } y \in M_i\}.$$

Let $\Omega = 2^{\Theta}$ and μ be an arbitrary probability measure on (Θ, Ω) .

PROPOSITION 4. Assume $\mathbb{E} = \{(\mathbb{Z}, \mathcal{P}) \mid \mathcal{P} \text{ is a partition of } \mathbb{Z}\}$, then the experimental paradigm $(\mathbb{Z}, (\Theta, \Omega, \mu), \mathbb{E})$ is concordant.

Proof of Proposition 4. First, observe that for any experiment $(A, \mathcal{P})$ and $P \in \mathcal{P}$, the set $\{\theta \mid c_{\theta} \in P\}$ is well defined and thus, (C1) holds. To show (C2) holds, take any arbitrary partition $\mathcal{W}$ of Θ and for each cell $W \in \mathcal{W}$, let

$$P_W = \left\{ \prod_{i=1}^{2^n-1} \{x \in M_i : x \succeq_{\theta} y \text{ for all } y \in M_i\} \mid \theta \in W \right\}.$$

Then, $\mathcal{P} = (P_W)_{W \in \mathcal{W}}$ is a partition of $\mathbb{Z}$ that induces the partition $\mathcal{W}$ of Θ . Hence, $(\mathbb{Z}, \mathcal{P})$ is the experiment described by (C2). ■

Notice that if this environment was augmented so that the subject made a single selection out of each decision problem (i.e., the modeler observed a choice function, rather than a choice correspondence) then concordance does not hold. Indeed, it would be impossible to distinguish between subjects who strictly prefer x to y and who are indifferent but break ties in favor of x . In such a case, the algebra Ω would have to be coarsened to accommodate this lack of observability. □

1.3 EXAMPLE: ADAPTIVE EXPERIMENTS

To illustrating the constraints imposed by (C2), we apply our framework to adaptive experimental design. Broadly, the literature on adaptive design argues that when facing constraints such as time, money, or subject fatigue, a modeler may wish to use a dynamic procedure in which the questions posed to a subject depend on the subject's earlier answers (see Drake et al. (2022) and Chapman et al. (2024)).

Example 3. Consider a modeler interested in a subject's preferences over some finite set X of cardinality bigger than or equal to 4, but who is constrained to ask the subject to choose from at most two binary menus. To maximize the information she can obtain, she used the following adaptive procedure: she first offers a binary menu $\{a, b\}$; if the subject chooses a , then she offers $\{c_a, d_a\} \subseteq X$, otherwise she offers $\{c_b, d_b\} \subseteq X$ where $\{c_a, d_a\}$ need not be equal $\{c_b, d_b\}$. Given the nature of the procedure, if the agent chooses a from $\{a, b\}$, then the modeler will know the choice out of $\{c_a, d_a\}$ but not the choice out of $\{c_b, d_b\}$, and analogously for a choice of b . Figure 1 illustrates the procedure.

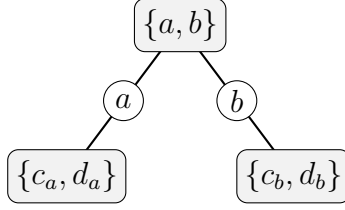


Figure 1: Adaptive Procedure

We capture the choice patterns generated by a two-shot adaptive procedure by taking $\mathbb{Z} = X \times X$. Each adaptive experiment is then a pair $(A, \mathcal{P})$ where

$$A = \{(a, c_a), (a, d_a), (b, c_b), (b, d_b)\}.$$

for some $a, c_a, d_a, b, c_b, d_b \in X$ and $\mathcal{P}$ is a partition of A .

The modeler's theory about the subject's level of sophistication is crucial in this setting. To illustrate, consider first a modeler who assumes the subject is *naive*, choosing her preferred option in the first round without accounting for how it affects the second-stage choice. This can be captured by taking Θ to be a set of utility functions over X and defining

$$c_\theta(A) = (x, y) \quad \text{such that} \quad x \in \arg \max_{z \in \{a, b\}} \theta(z) \quad \text{and} \quad y \in \arg \max_{w \in \{c_x, d_x\}} \theta(w).$$

Next, consider a modeler who believes the subject is *sophisticated*. In this case, types are associated with the following choice behavior:

$$c_\theta(A) = (x, y) \in \arg \max_{(z, w) \in A} (\theta(z) + \theta(w)).$$

Finally, the modeler may wish to allow for both naive and sophisticated subjects, in which case Θ can encode not only the subject's utility function but also her level of sophistication.

In any case, the set of experiments $\mathbb{E}$ can only test a very limited set of hypotheses, and so in general (C2) will not hold. This is because, for any experiment in $\mathbb{E}$, there is at most four observable outcomes, and thus, precluding the identification of any finer partitions. For instance, the experiment predicated on $A = \{(a, c_a), (a, d_a), (b, c_b), (b, d_b)\}$, assuming all choice objects are distinct, cannot simultaneously determine a subject's preference between c_a and d_a and also about c_b and d_b . Notice that deeper trees of choices and highly constrained initial beliefs μ both offer a solution to this observability problem. $\square$

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